

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. The region in the first quadrant between the x axis and $y = 6x - x^2$ is rotated about the y axis. The volume of this solid of revolution is **1**

(A) $\int_0^6 \pi (6x - x^2) \, dx$

(C) $\int_0^6 \pi x (6x - x^2)^2 \, dx$

(B) $\int_0^6 2\pi x (6x - x^2) \, dx$

(D) $\int_0^6 \pi \left(3 + \sqrt{9 - y}\right)^2 \, dx$

2. What are all the values of k for which the graph of $y = x^3 - 3x^2 + k$ will have three distinct x intercepts? **1**

(A) all $k > 0$

(C) $k = 0, 4$

(B) all $k < 4$

(D) $0 < k < 4$

3. Which of the following is the triple root of the equation **1**

$$8x^4 + 12x^3 - 30x^2 + 17x - 3 = 0$$

(A) $\frac{1}{2}$

(B) $-\frac{5}{4}$

(C) -3

(D) 0

4. If n is a non-negative integer, then for what values of n is $\int_0^1 x^n \, dx = \int_0^1 (1-x)^n \, dx$ true? **1**

(A) no solution

(C) non zero n , only

(B) n even, only

(D) all values of n

5. What are the coordinates of the foci of $xy = 18$? **1**

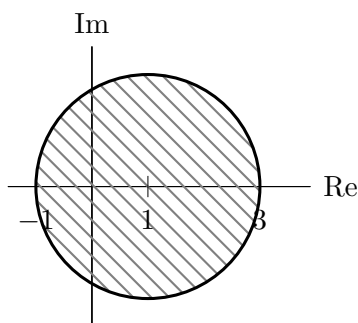
(A) $(0, 6), (0, -6)$

(C) $(3\sqrt{2}, 3\sqrt{2}), (-3\sqrt{2}, -3\sqrt{2})$

(B) $(0, 3\sqrt{2}), (0, -3\sqrt{2})$

(D) $(6, 6), (-6, 6)$

6. Which of the following inequalities is represented by the Argand diagram? 1



- (A) $|z - 1| \leq 2$ (C) $|z + 1| \leq 2$
 (B) $|z - i| \leq 2$ (D) $|z + i| \leq 2$
7. What does $\int \frac{dx}{(x-1)(x+2)}$ evaluate to? 1
 (A) $\frac{1}{3} \log_e \left| \frac{x-1}{x+2} \right| + C$ (C) $\frac{1}{3} \log_e |(x-1)(x+2)| + C$
 (B) $\frac{1}{3} \log_e \left| \frac{x+2}{x-1} \right| + C$ (D) $(\log_e |x-1|)(\log_e |x+2|)$
8. What is the value of $\int_0^1 x e^{-x} dx$? 1
 (A) $1 - 2e$ (C) $1 - 2e^{-1}$
 (B) -1 (D) $2e - 1$
9. What is the value of the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$? 1
 (A) $\frac{3}{\sqrt{13}}$ (C) $\frac{11}{\sqrt{5}}$
 (B) $\frac{\sqrt{13}}{3}$ (D) $\sqrt{2}$
10. What is the value of $\frac{dy}{dx}$ at the point $(1, 2)$ if $xy^2 + 2xy = 8$? 1
 (A) $-\frac{5}{2}$ (C) -1
 (B) $-\frac{4}{3}$ (D) $-\frac{1}{2}$

End of Section I
Examination continues overleaf...

Section II: Short answer

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Evaluate:		
i. $\int \frac{dx}{\sqrt{7-9x-x^2}}$		2
ii. $\int \frac{dx}{x \log_e x}$		2
(b) Evaluate $\int_1^2 \frac{dx}{x(1+x^2)}$		4
(c) Evaluate $\int \frac{x}{\sqrt{1-x}} dx$		3
(d) Find $\int e^{-2x} \cos x dx$		4

Question 12 (15 Marks)	Commence a NEW page.	Marks
(a) Show that $3i$ is a root of $P(x) = x^4 - 3x^3 + 5x^2 - 27x - 36$, and hence solve $P(x) = 0$ completely.		3
(b) If α , β and γ are roots of $3x^3 + 4x^2 + 5x + 1 = 0$, find the value of $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$		3
(c) Given $Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$ has a root of multiplicity 2, solve $Q(x) = 0$ over $\mathbb{C}$.		3
(d) The roots of the polynomial equation $x^3 - 2x^2 + 3x + 1 = 0$ are α , β and γ . Find the value of $\alpha^3 + \beta^3 + \gamma^3$.		3
(e) The polynomial $x^5 - ax^2 + b = 0$ has a multiple root. Show that $108a^5 = 3125b^3$.		3

Question 13 (15 Marks)

Commence a NEW page.

Marks

- (a) Sketch the region in the Argand diagram which simultaneously satisfies the following inequalities: **2**

$$\begin{cases} |z - 2i| \leq 2 \\ \operatorname{Im}(z) \geq 2 \end{cases}$$

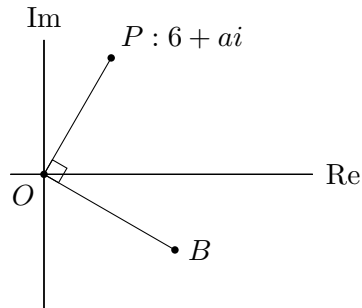
- (b) What is the locus in the Argand diagram of the point z such that **3**

$$z\bar{z} - 2(z + \bar{z}) = 5$$

- (c) Find the value of z^{10} in Cartesian form, given that **3**

$$z = \sqrt{2} - \sqrt{2}i$$

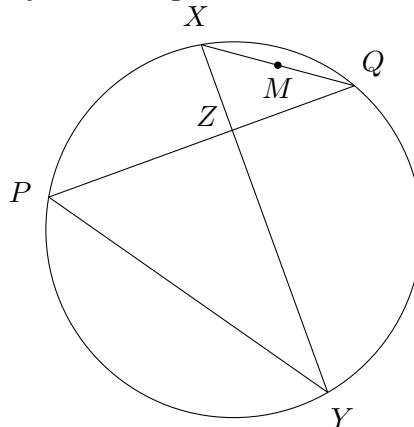
- (d) In the following Argand diagram, P represents the point $6 + ai$, and O is the origin. **3**



Find the complex number represented by the point B , given $\angle POB = 90^\circ$ and

$$2|OB| = 3|OP|$$

- (e) Two perpendicular chords PQ and XY of a circle intersect at Z . Copy the diagram into your writing booklet. **4**



If M is the midpoint of the chord QX , prove that MZ produced is perpendicular to the chord PY .

Question 14 (15 Marks)

Commence a NEW page.

Marks

(a) Sketch the following graphs:

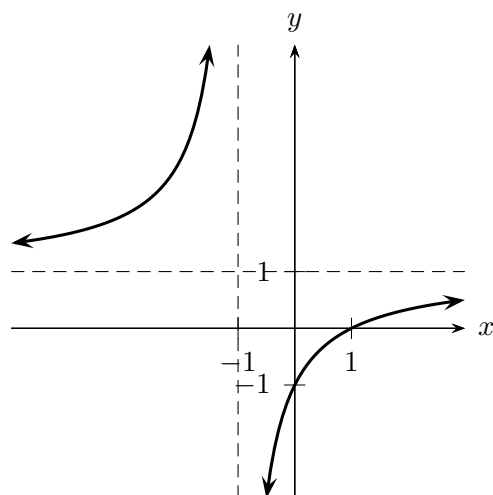
i. $y = |\sin x|$ for $-2\pi \leq x \leq 2\pi$.

1

ii. $y = \sqrt{x^2 - 4}$

2

iii. $y^2 = x^2 - 9x$

2(b) Sketch $y = \frac{1}{(x-1)^2(x+2)}$.**2**(c) The diagram shows the graph of $f(x)$.

Sketch the following curves on separate diagrams, clearly indicating any turning points and asymptotes.

i. $y = \frac{1}{f(x)}$

2

ii. $y = f(|x|)$

2

iii. $y = \log_e(f(x))$

2

iv. $y = e^{f(x)}$

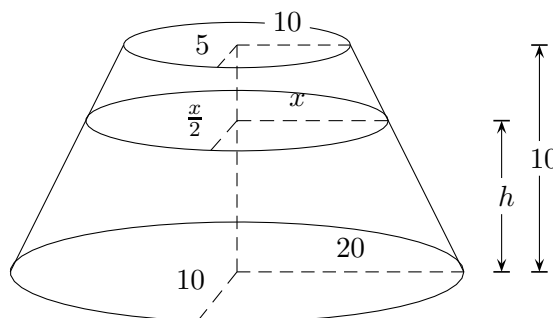
2

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) A solid of height 10 m stands on horizontal ground. The base of the solid is an ellipse with semi-axes of 20 m and 10 m. Horizontal cross-sections taken parallel to the base and at height h metres above the base are ellipses with semi-axes x metres and $\frac{x}{2}$ metres so that the centres of these elliptical cross-sections lie on a vertical straight line, and the extremities of their semi-axes lie on sloping straight lines as shown in the diagram. The top of the solid is an ellipse with semi-axes 10 m and 5 m. **5**

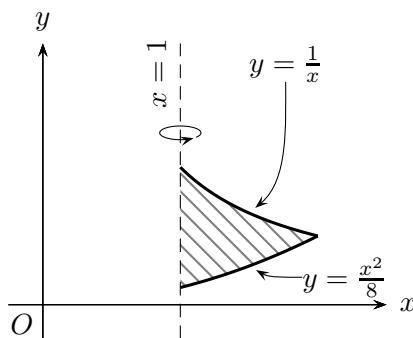


Show that the volume $V \text{ m}^3$ of the solid is given by

$$V = \frac{\pi}{2} \int_0^{10} (20 - h)^2 dh$$

and hence find the volume correct to the nearest cubic metre.

- (b) The shaded region shown in the diagram below is bounded by $y = \frac{1}{x}$, $y = \frac{x^2}{8}$ and $x = 1$. This region is rotated about the line $x = 1$.



- i. Find an integral which gives the volume of the resulting solid of revolution using the method of cylindrical shells. **4**
 - ii. Find the volume of the solid of revolution. **2**
- (c) On the number plane, shade the region **4**

$$(x - a)^2 + (y - b)^2 \leq R^2$$

where $R < b < a$.

Find the volume when this shape is rotated about the y axis using the method of slices.

Question 16 (15 Marks) Commence a NEW page. **Marks**

- (a) i. Determine the real values of p for which the equation

$$\frac{x^2}{3+p} + \frac{y^2}{8+p} = 1$$

defines

(α) an ellipse **1**

(β) a hyperbola **2**

- ii. For the value $p = -4$ in the above equation, find the **2**

- eccentricity
- coordinates of the foci, and
- the equations of the directrices

of the conic.

- (b) P is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre at the origin O .

A line drawn from the origin O , parallel to the tangent to the ellipse at P , meets the ellipse at Q .

- i. Derive the equation of the tangent at $P(a \cos \theta, b \sin \theta)$. **2**

- ii. Hence or otherwise, prove that the area of $\triangle OPQ$ is independent of the position of P . **3**

- (c) i. Find the equation of the normal at $P(a \sec \theta, b \tan \theta)$ to the hyperbola **2**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

- ii. This normal intersects the x and y axes at Q and R respectively. $M(x, y)$ is the midpoint of QR . Find the equation of the locus of M as P varies on the hyperbola. **3**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Mr Weiss

☐ 12M4B – Mr Ireland

☐ 12M4C – Mr Fletcher

- | | | | | | |
|-----------|---|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 | – | ☐ A | ☐ B | ☐ C | ☐ D |

2012 Ext. 2 - Test 3

iii) a) i/ $\int \frac{dx}{\sqrt{7-9x-x^2}}$

$$= \int \frac{dx}{\sqrt{\frac{109}{4} - \left(\frac{9}{2} - x\right)^2}}$$

$$= \sin^{-1} \left(\frac{2x+9}{\sqrt{109}} \right) + C$$

ii/ $\int \frac{\frac{1}{x} dx}{\ln x}$

$$= \ln(\ln|x|) + C$$

b) $\int_1^2 \frac{dx}{x(1+x^2)}$

Partial fractions

$$\frac{1}{x(1+x^2)} = \frac{a}{x} + \frac{bx+c}{1+x^2}$$

$$1 = a(1+x^2) + bx^2 + c$$

$$\therefore a+b=0$$

$$c=0$$

$$a=1 \quad b=-1$$

$$\int_1^2 \frac{dx}{x(1+x^2)} = \int_1^2 \left(\frac{1}{x} - \frac{x}{1+x^2} \right) dx$$

$$= \ln x - \frac{1}{2} \ln(1+x^2) \Big|_1^2$$

$$= \ln 2 - \ln 1 - \frac{1}{2} \ln 5 + \frac{1}{2} \ln 2$$

$$= \frac{3}{2} \ln 2 - \frac{1}{2} \ln 5$$

$$= \ln 2 \sqrt{\frac{2}{5}}$$

$$\approx 0.235 \text{ to 3 D.P.}$$

c) $\int \frac{x dx}{\sqrt{1-x}}$

$$u^2 = 1-x$$

$$2u du = -dx$$

$$\therefore \int \left(\frac{1-u^2}{u} \right) \cdot -2u du$$

$$= \int -\frac{2u}{u} + \frac{2u^3}{u} du$$

$$= \frac{2u^3}{3} - 2u + C$$

$$= \frac{2}{3} (1-x)^{3/2} - 2(1-x)^{1/2} + C$$

d) $\int \frac{u}{e^{-2u}} \cos u du$

$$I = e^{-2u} \sin u + 2 \int \sin u e^{-2u} du$$

$$= e^{-2u} \sin u - 2 \cos u e^{-2u} - 4 \int \cos u e^{-2u} du$$

$$\therefore 5I = e^{-2u} \sin u - 2 \cos u e^{-2u}$$

$$= \frac{1}{5} e^{-2u} (\sin u - 2 \cos u) + C$$

Q1 - Q10

- | | |
|------|-------|
| 1. B | 6. A |
| 2. D | 7. A |
| 3. A | 8. C |
| 4. D | 9. B |
| 5. D | 10. B |

Q12

(a) $P(x) = x^4 - 3x^3 + 5x^2 - 27x - 36$

$$P(3i) = (3i)^4 - 3(3i)^3 + 5(3i)^2 - 27(3i) - 36$$

$$= 81 + 81i - 45 - 81i - 36$$

$$= 0$$

$\therefore 3i$ is a root ✓

Real coeff $\therefore -3i$ also a root.

$$P(x) = (x+3i)(x-3i)Q(x)$$

$$= (x^2+9)(x^2-3x-4)$$

$$= (x^2+9)(x-4)(x+1)$$

$\therefore$ roots are $\pm 3i, 4, -1$ ✓✓

(ALT. use sum & product of roots).

(b) $3x^3 + 4x^2 + 5x + 1 = 0$

$$\frac{1}{\alpha^2\beta^2} + \frac{1}{\alpha^2\gamma^2} + \frac{1}{\beta^2\gamma^2} = \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha\beta\gamma)^2}$$

$$= \frac{(\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{\left(-\frac{4}{3}\right)^2 - 2\left(\frac{5}{3}\right)}{\left(-\frac{1}{3}\right)^2}$$

$$= -14$$

(ALT: Create the equation with roots equal to α^2, β^2 and γ^2 , and proceed from there.)

Q12

(c) $Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$

$$Q'(x) = 4x^3 - 15x^2 + 8x + 3$$

First, find roots of $Q'(x) = 0$.

$$Q'(1) = 0 \text{ but } Q(1) \neq 0$$

$$Q'(3) = 0, \text{ and } Q(3) = 0 \therefore x=3 \text{ is the double root.} \quad \checkmark$$

$$\therefore Q(x) = (x-3)^2 \cdot S(x)$$

$$= (x-3)^2 (x^2+x+1) \quad (\text{by inspection or otherwise}) \quad \checkmark$$

$$\therefore (x-3)^2 = 0 \text{ or } x^2+x+1 = 0$$

$$\therefore x = 3, 3, \frac{-1 \pm i\sqrt{3}}{2} \quad \checkmark$$

(d) $P(x) = x^3 - 2x^2 + 3x + 1 = 0$

α, β, γ are roots, $\therefore$

$$\alpha^3 - 2\alpha^2 + 3\alpha + 1 = 0$$

$$\beta^3 - 2\beta^2 + 3\beta + 1 = 0$$

$$\gamma^3 - 2\gamma^2 + 3\gamma + 1 = 0$$

adding:

$$\alpha^3 + \beta^3 + \gamma^3 - 2(\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha + \beta + \gamma) + 3 = 0$$

✓ (method).

$$\text{Now } \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= (2)^2 - 2\left(\frac{3}{2}\right) = -2 \quad \checkmark$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 - 2(-2) + 3(2) + 3 = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -13 \quad \checkmark \text{ (final answer)}$$

Q12

(e)

$P(x) = x^5 - ax^2 + b = 0$ has a multiple root.

call it α .

Now $P'(x) = 5x^4 - 2ax$.

We have $P(\alpha) = P'(\alpha) = 0$

$$\therefore \alpha^5 - a\alpha^2 + b = 0 \quad \dots (1)$$

$$5\alpha^4 - 2a\alpha = 0 \quad \dots (2)$$

From (2), $\alpha(5\alpha^3 - 2a) = 0$

$$\therefore \alpha = 0 \quad \text{or} \quad \alpha^3 = \frac{2a}{5}$$

But $P(0) \neq 0 \therefore \alpha^3 = \frac{2a}{5} \therefore \alpha = \left(\frac{2a}{5}\right)^{\frac{1}{3}}$ ✓

Sub. into (1):

$$\left(\frac{2a}{5}\right)^{\frac{5}{3}} - a \cdot \left(\frac{2a}{5}\right)^{\frac{2}{3}} = -b$$
 ✓

Factorise:-

$$a^{\frac{5}{3}} \cdot \left(\frac{2}{5}\right)^{\frac{2}{3}} \left(\frac{2}{5} - 1\right) = -b$$

$$a^{\frac{5}{3}} \cdot \left(\frac{2}{5}\right)^{\frac{2}{3}} \cdot \left(-\frac{3}{5}\right) = -b$$

Cube both sides:

$$a^5 \left(\frac{2}{5}\right)^2 \left(-\frac{3}{5}\right)^3 = -b^3$$

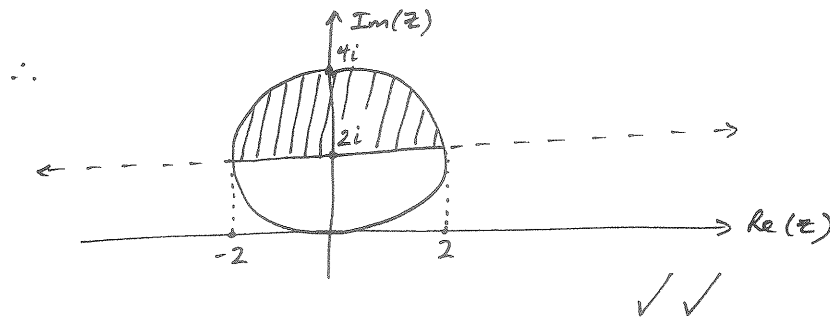
$$\frac{-108 a^5}{3125} = -b^3$$

$$\therefore 108 a^5 = 3125 b^3$$
 ✓

[Note: Failure to eliminate $\alpha = 0$ as a possibility costs a mark.]

Q13

(a) $|z - 2i| \leq 2$ and $\text{Im}(z) \geq 2$



(b) $z\bar{z} - 2(z + \bar{z}) = 5$

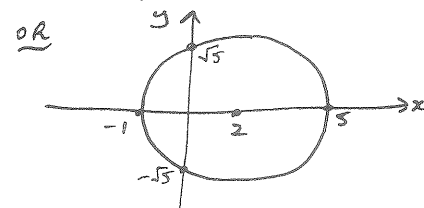
Let $z = x + iy$ $\therefore \bar{z} = x - iy$
 $\therefore z\bar{z} = x^2 + y^2, \quad z + \bar{z} = 2x$ ✓

$$\therefore x^2 + y^2 - 4x = 5$$

$$x^2 - 4x + 4 + y^2 = 9$$

$$(x-2)^2 + y^2 = 3^2$$
 ✓

This is a circle, centre $(2, 0)$, radius 3



[Note: full marks requires a description of the locus or a sketch depicting it.]

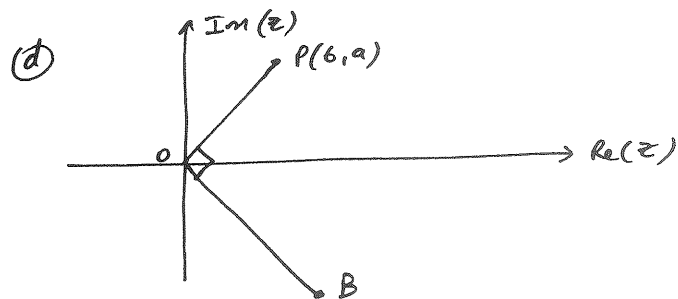
Q13

$$\begin{aligned} \textcircled{c} \quad z &= \sqrt{2} - \sqrt{2}i \\ &= 2 \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right) \\ &= 2 \operatorname{cis} \left(-\frac{\pi}{4} \right) \end{aligned}$$

$$\therefore z^{10} = 2^{10} \cdot \operatorname{cis} \left(-\frac{5\pi}{2} \right)$$

(by De Moivre)

$$\begin{aligned} &= 2^{10} \operatorname{cis} \left(-\frac{\pi}{2} \right) = 1024 \cdot -i \\ &= -1024i \end{aligned}$$



We're told $|\vec{OB}| = \frac{3}{2} \cdot |\vec{OP}|$

Thus $\vec{OB}$ is obtained from $\vec{OP}$ by a clockwise rotation of $\frac{\pi}{2}$ and a stretching by a factor of $\frac{3}{2}$.

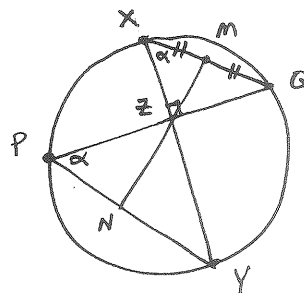
$$\begin{aligned} \therefore \vec{OB} \text{ represents } & \frac{3}{2} \cdot -i \cdot (6+ai) \\ &= -9i + \frac{3a}{2} \end{aligned}$$

$$\therefore B \text{ represents } \frac{3a}{2} - 9i.$$

✓ (final answer)

Q13

(e)



Given: $PQ \perp XY$
 $MX = MQ$

Let MZ produced meet PY at N .

To prove: $MN \perp PY$

Since $\angle XZQ = 90^\circ$ and $MX = MQ$,
 $\therefore XQ$ is the diameter of a circle, centre M ,
that passes through Z . [converse to angle in a semicircle is 90°]
Thus $MZ = MX = MQ$

Let $\angle YPQ = \alpha$
 $\therefore \angle ZXQ = \alpha$ (angles standing on same arc QY)
 $\therefore \angle XQZ = 90 - \alpha$ (angle sum of $\triangle XZQ$, given $\angle XZQ = 90^\circ$)

Now $\angle MZQ = \angle MQZ$ (base angles of isosceles $\triangle MZQ$)
 $= 90 - \alpha$

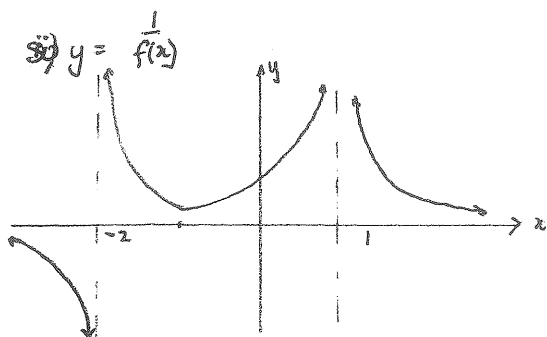
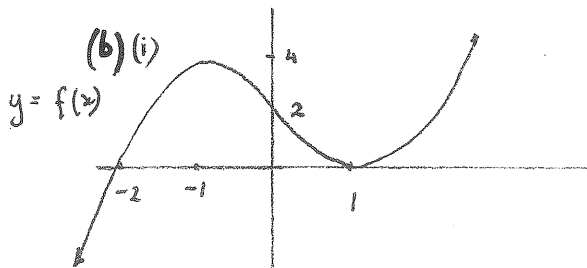
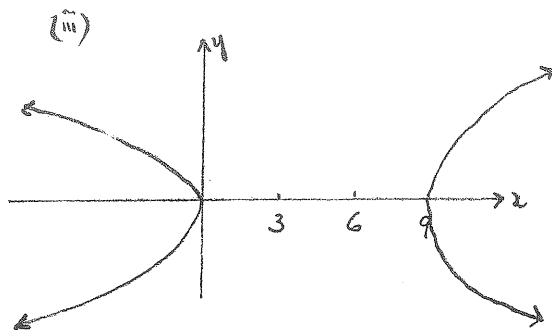
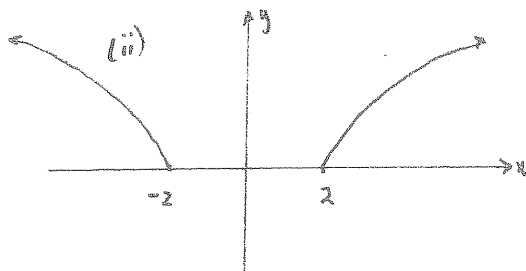
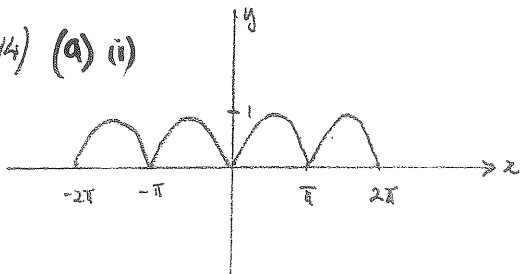
Thus $\angle PZN = \angle MZQ$ (vertically opposite $\angle$ s)
 $= 90 - \alpha$

Thus $\angle PNZ = 180 - \alpha - (90 - \alpha)$ [angle sum of $\triangle PZN$]
 $= 90^\circ$

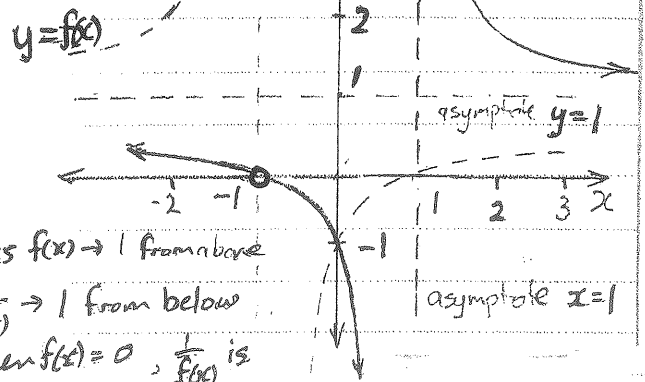
That is, $MN \perp PY$. #

[Note: no marks awarded without a viable strategy towards solution being supplied. eg. 'angles on same arc' by itself gets no marks.]

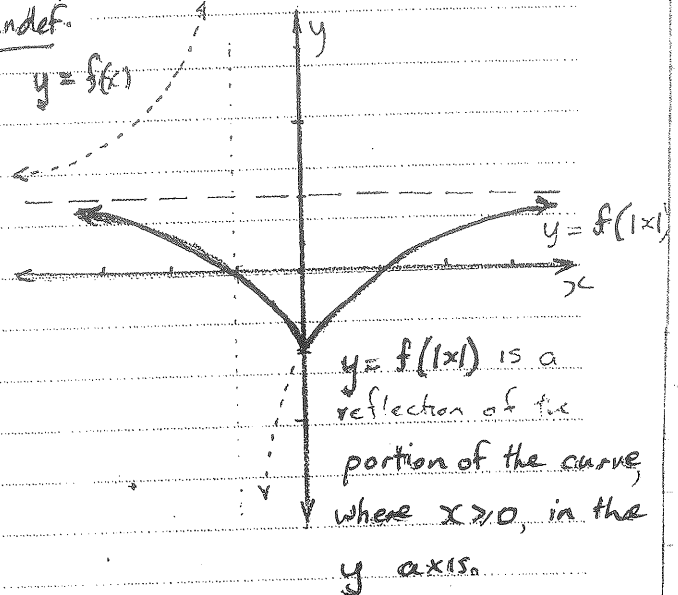
Q 14) (a) (i)



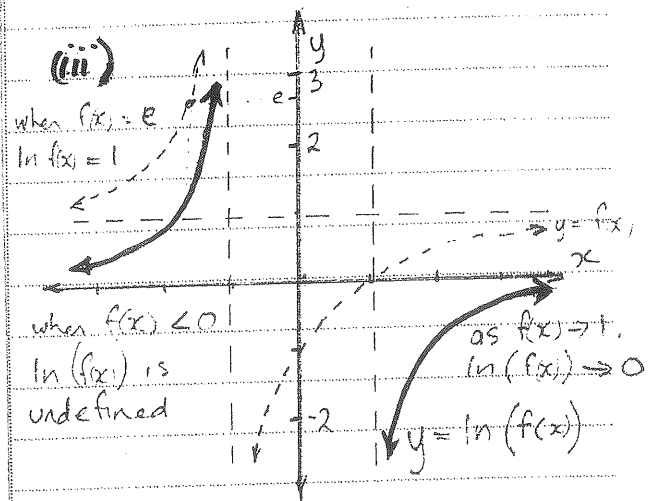
(c) (i)



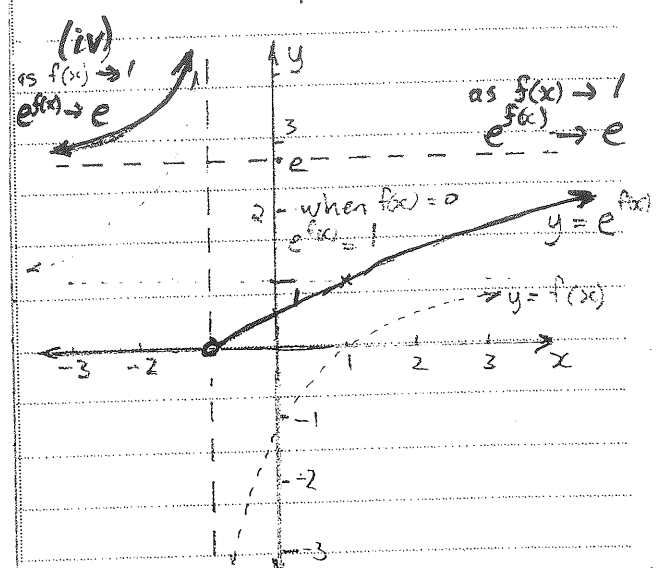
(ii)



(iii)

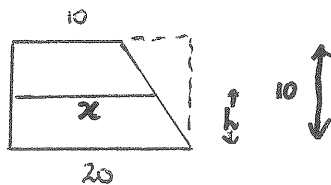


(iv)



(Q15)

a)



Using similar triangles

$$\frac{20-x}{10} = \frac{h}{10}$$

$$x = 20-h$$

$$\begin{aligned} \text{Area of slice } A &= \pi \times \frac{x}{2} \\ &= \frac{\pi}{2} (20-h)^2 \end{aligned}$$

Volume of solid

$$V = \lim_{\delta h \rightarrow 0} \sum_{h=0}^{10} \frac{\pi}{2} (20-h)^2 \delta h$$

$$V = \frac{\pi}{2} \int_0^{10} (20-h)^2 dh$$

$$= -\frac{\pi}{6} (20-h)^3 \Big|_0^{10}$$

$$= -\frac{\pi}{6} (10^3 - 20^3)$$

$$= \frac{7000\pi}{6} \text{ m}^3$$

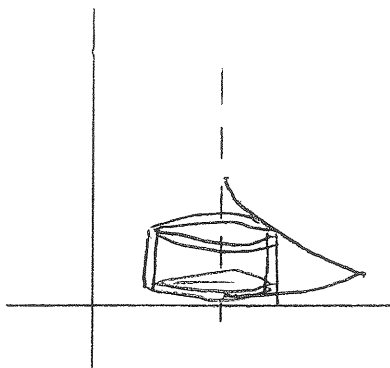
$$= 3665 \text{ m}^3 \text{ (nearest m}^3\text{)}$$

b)

$$y = \frac{1}{x} \text{ and } y = \frac{x^2}{8}$$

$$\therefore x^3 = 8$$

$$x = 2$$



$$V = \lim_{\delta x \rightarrow 0} \sum_{x=1}^2 2\pi (x-1) \left(\frac{1}{x} - \frac{x^2}{8} \right) \delta x$$

$$= 2\pi \int_1^2 (x-1) \left(\frac{1}{x} - \frac{x^2}{8} \right) dx$$

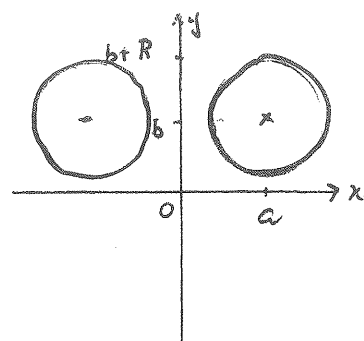
$$= 2\pi \int_1^2 \left(1 - \frac{x^3}{8} - \frac{1}{x} + \frac{x^2}{8} \right) dx$$

$$= 2\pi \left[x - \frac{x^4}{32} - \ln x + \frac{x^3}{24} \right]_1^2$$

$$= 2\pi \left(\frac{79}{96} - \ln 2 \right)$$

$$= \frac{79\pi}{48} - 2\pi \ln 2$$

c)



Rotation about y-axis

$$\text{Area} = \pi (R^2 - r^2)$$

$$\text{Now } R = a + \sqrt{R^2 - (y-b)^2}$$

$$r = a - \sqrt{R^2 - (y-b)^2}$$

$$V_{\text{slice}} = \pi (R+r)(R-r) \delta y$$

$$V_{\text{slice}} = \pi (2a) \left(2 \sqrt{R^2 - (y-b)^2} \right) \delta y$$

$$V = 4\pi a \int_{b-R}^{b+R} \sqrt{R^2 - (y-b)^2} dy$$

$$= 4\pi a \cdot \frac{1}{2} \pi R^2 \quad \begin{array}{l} \text{semi-circle} \\ \text{rad } R, \text{ centre } b. \end{array}$$

$$= 2\pi^2 R^2 a \text{ u}^3$$

Q16) (a) $\frac{x^2}{3+p} + \frac{y^2}{8+p} = 1$

(i) α) ellipse if $3+p > 0$
and $8+p > 0$
 $\therefore p > -3$

β) hyperbola if $3+p > 0$
and $8+p < 0$
 $\therefore$ Not possible
or if $3+p < 0$ and $8+p > 0$

$\therefore -8 < p < -3$

(ii) if $p = -4$
 $\frac{x^2}{-1} + \frac{y^2}{4} = 1$
 $\frac{y^2}{4} - \frac{x^2}{1} = 1$

Now $b^2 = a^2(e^2 - 1)$
 $1 = 4(e^2 - 1)$
 $e = \pm \frac{\sqrt{5}}{2}$

foci $(0, \pm \sqrt{5})$

directrices $y = \pm \frac{4}{\sqrt{5}}$

vertices $(0, \pm 2)$

(b) (i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$
 $= -\frac{b^2 x \cos \theta}{a^2 b \sin \theta}$

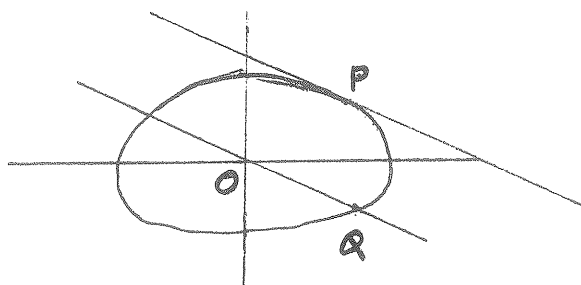
eqn of tangent

$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$
 $bx \cos \theta + ay \sin \theta = ab (\sin^2 \theta + \cos^2 \theta)$

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

(ii)



eqn of OQ $y = \frac{-b \cos \theta x}{a \sin \theta}$ ①

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ②

$\frac{x^2}{a^2} + \frac{1}{b^2} \frac{b^2 \cos^2 \theta x^2}{a^2 \sin^2 \theta} = 1$

$\sin^2 \theta x^2 + \cos^2 \theta x^2 = a^2 \sin^2 \theta$

$\therefore x^2 = a^2 \sin^2 \theta$

$x = \pm a \sin \theta$

from diagram $x = a \sin \theta$

$\therefore y = -b \cos \theta$

Dist from P to OQ

OQ $b \cos \theta x + a \sin \theta y = 0$

P $(a \cos \theta, b \sin \theta)$

$\therefore$ Dist $= \frac{ab \cos^2 \theta + ab \sin^2 \theta}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$

$= \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$

dist OQ $= \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$

Δ Area $= \frac{1}{2} \cdot OQ \cdot \text{dist from P}$
 $= \frac{1}{2} \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \times \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$
 $= \frac{ab}{2}$ indep of P.

Q16-continued)

c) eqn. of normal at P

$$(i) \quad ax \sin \theta + by = (a^2 + b^2) \tan \theta$$

ii) at Q $y=0$

$$\therefore x = \frac{(a^2 + b^2) \tan \theta}{a \sin \theta}$$

$$= \frac{(a^2 + b^2) \sec \theta}{a}$$

at R $x=0$

$$y = \frac{a^2 + b^2}{b} \tan \theta$$

at M midpoint of RQ

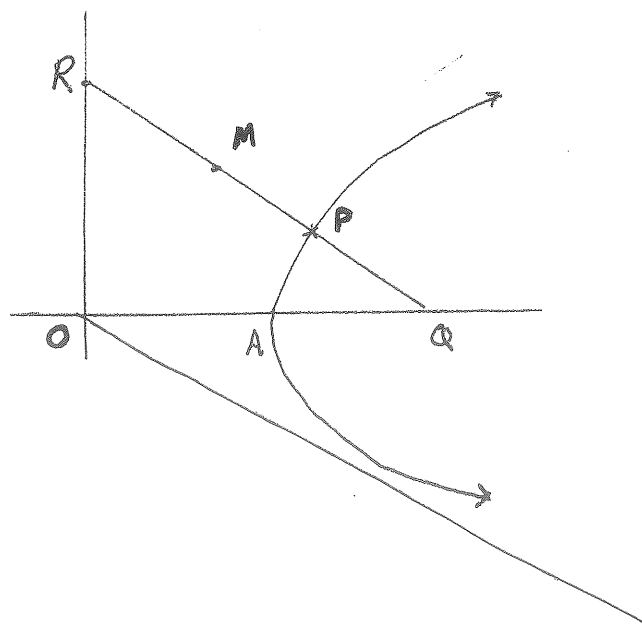
$$X = \frac{1}{2a} (a^2 + b^2) \sec \theta \quad Y = \frac{1}{2b} (a^2 + b^2) \tan \theta$$

locus of M

$$(2aX)^2 - (2bY)^2 = (a^2 + b^2)^2 (\sec^2 \theta - \tan^2 \theta)$$

$$\therefore 4a^2 X^2 - 4b^2 Y^2 = (a^2 + b^2)^2$$

which is another hyperbola.



Derivation of c)i/

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x b^2}{y a^2}$$

$$= \frac{b^2 x \sec \theta}{a^2 b \tan \theta}$$

eqn of normal $y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$

$$ax \tan \theta + by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$$

$$(\div \sec \theta) \quad ax \sin \theta + by = (a^2 + b^2) \tan \theta$$

$$\text{OR} \quad \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

$$(\div \sec \theta \tan \theta) \quad \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$